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Experiment 3

Aim: To deploy multi-service applications using Docker Compose.

Lab Outcome: Mapped LO4

LO4: Containerize, orchestrate and deploy applications using Docker, Docker Compose, Kubernetes and cloud infrastructure tools.

Theory

Docker Compose

Docker Compose is a utility that helps manage applications made up of multiple Docker containers. Rather than creating and running each container separately, it allows all required containers to be defined in a single compose.yml file. This file specifies the application's services, networking, storage, and other configuration details.

Using Docker Compose, all the containers required by an application can be started, stopped, and managed together through a single command.

Docker Compose is widely used for applications that require multiple components, for example:

- Web Server: Nginx or Apache
- Database: MySQL or PostgreSQL
- Cache: Redis
- Application Server: Node.js, Python, or Java-based applications

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Docker Compose Configuration File

The application configuration is stored in a file named compose.yml, which defines all the services and resources required for deployment.

The configuration includes:

- Web Service: Nginx web server for serving web content.
- Database Service: MySQL 8.0 database container.
- Volume: A persistent storage volume used to retain MySQL data even if the container is removed.
- Network: An automatically generated Docker network that enables communication between the defined services.

Benefits of Docker Compose

- Starts and manages multiple containers using a single command.
- Keeps application configuration organized within one YAML file.
- Automatically creates an isolated network for inter-container communication.
- Supports persistent data storage using Docker volumes.
- Improves portability by allowing the same application setup across different systems.
- Makes development, testing, and deployment easier and more efficient.

Docker Compose Commands

Command	Description
docker compose up -d	Builds (if required) and launches all services defined in the compose.yml file in detached (background) mode.
docker compose ps	Lists all containers managed by Docker Compose along with their current status.
docker compose logs	Displays log messages generated by the running services for monitoring and troubleshooting.
docker compose down	Stops and removes all containers, networks, and other resources created for the application.
docker compose down -v	Stops the application and removes all associated containers, networks, and volumes, including any stored persistent data.

Output:

Welcome compose.yml

C: > Users > 15L > Desktop > compose.yml

```yaml
version: "3.9"

services:
  web:
    image: nginx:latest
    ports:
      - "8081:80"
    depends_on:
      - db

  db:
    image: mysql:8.0
    environment:
      MYSQL_ROOT_PASSWORD: mypassword
      MYSQL_DATABASE: mydb
    volumes:
      - db_data:/var/lib/mysql

volumes:
  db_data:
```

```
Microsoft Windows [Version 10.0.26200.8894]
(c) Microsoft Corporation. All rights reserved.

C:\Users\15L>cd
C:\Users\15L

C:\Users\15L>cd desktop

C:\Users\15L\Desktop>dir
 Volume in drive C is Windows
 Volume Serial Number is 1CB8-42B2

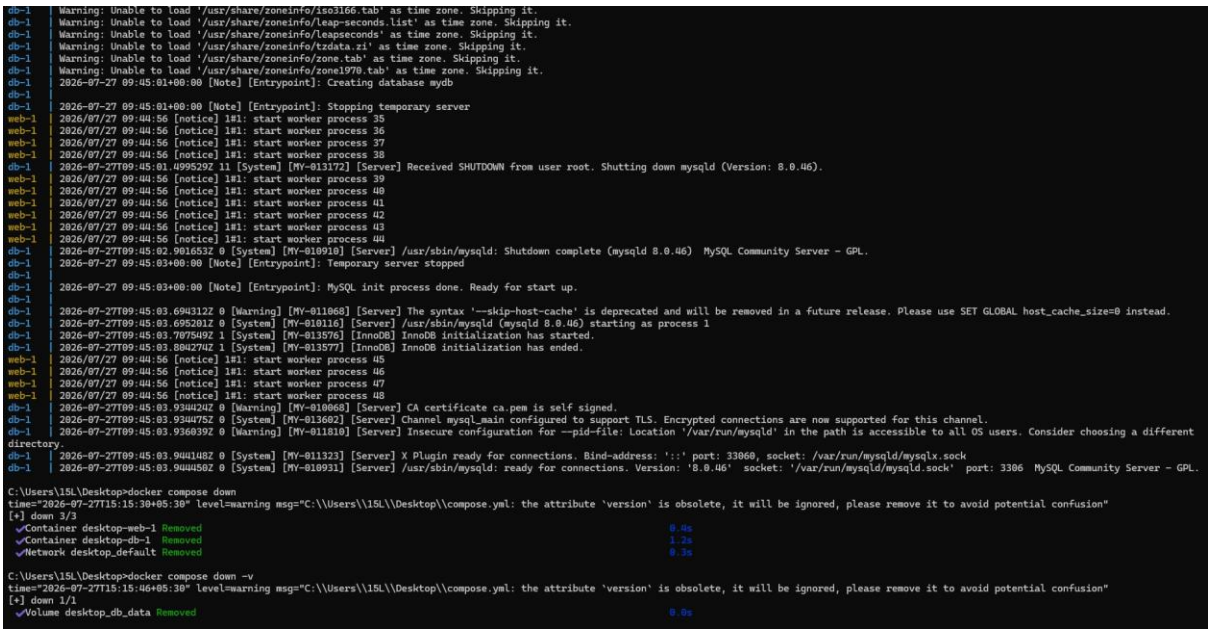
 Directory of C:\Users\15L\Desktop

07/27/2026  03:04 PM    <DIR>          .
07/20/2026  04:36 PM    <DIR>          ..
11/01/2025  10:27 AM    <DIR>          .dist
10/04/2023  09:23 AM             1,095 Cisco Packet Tracer.lnk
10/25/2025  10:37 AM               956 CodeBlocks.lnk
07/27/2026  03:13 PM               306 compose.yml
08/12/2024  01:11 PM               699 Desktop.lnk
03/05/2026  01:53 PM             2,143 Docker Desktop.lnk
04/18/2023  10:10 AM             1,095 Eclipse IDE for Java Developers - 2023-03.lnk
08/07/2025  03:42 PM             1,095 Eclipse IDE for Java Developers - 2024-12.lnk
11/03/2025  08:37 AM             1,287 GIMP 3.0.6-1.lnk
07/14/2026  10:24 AM    <DIR>          Git
10/26/2023  12:16 PM               751 GNU Prolog.lnk
07/16/2026  08:21 AM    <DIR>          html
01/10/2026  11:57 AM             2,357 Microsoft Edge.lnk
07/21/2026  11:16 AM           328,819 Miheeka_Exp02_T12_ASD&D.pdf
04/24/2023  11:30 AM               886 MinGW Installer.lnk
09/19/2024  01:23 PM               660 TurboC++ for Windows 7.lnk
10/26/2023  12:17 PM             1,401 Visual Studio Code.lnk
              14 File(s)        343,550 bytes
               5 Dir(s)  699,451,424,768 bytes free
```

```
C:\Users\15L\Desktop>docker compose up -d
time="2026-07-27T15:14:53+05:30" level=warning msg="C:\\Users\\15L\\Desktop\\compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] up 4/4
 ✔Network desktop_default Created                                                    0.0s
 ✔Volume desktop_db_data  Created                                                    0.0s
 ✔Container desktop-db-1  Created                                                    0.0s
 ✔Container desktop-web-1 Created                                                    0.1s

C:\Users\15L\Desktop>docker compose ps
time="2026-07-27T15:15:09+05:30" level=warning msg="C:\\Users\\15L\\Desktop\\compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
NAME           IMAGE          COMMAND                  SERVICE   CREATED         STATUS         PORTS
desktop-db-1   mysql:8.0      "docker-entrypoint.s…"   db        13 seconds ago   Up 14 seconds   3306/tcp, 33060/tcp
desktop-web-1  nginx:latest   "/docker-entrypoint.…"   web       13 seconds ago   Up 14 seconds   0.0.0.0:8081->80/tcp, [::]:8081->80/tcp
```

```
C:\Users\15L\Desktop>docker compose logs
time="2026-07-27T15:15:20+05:30" level=warning msg="C:\\Users\\15L\\Desktop\\compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
db-1  | 2026-07-27 09:44:56+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.46-1.el9 started.
db-1  | 2026-07-27 09:44:56+00:00 [Note] [Entrypoint]: Switching to dedicated user 'mysql'
db-1  | 2026-07-27 09:44:56+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.46-1.el9 started.
db-1  | 2026-07-27 09:44:56+00:00 [Note] [Entrypoint]: Initializing database files
web-1 | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
web-1 | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
web-1 | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
db-1  | 2026-07-27T09:44:56.446077Z 0 [Warning] [MY-011068] [Server] The syntax '--skip-host-cache' is deprecated and will be removed in a future release. Please use SET GLOBAL host_cache_size=0 instead.
db-1  | 2026-07-27T09:44:56.446133Z 0 [System] [MY-013169] [Server] /usr/sbin/mysqld (mysqld 8.0.46) initializing of server in progress as process 80
db-1  | 2026-07-27T09:44:56.450557Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
db-1  | 2026-07-27T09:44:56.703658Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
db-1  | 2026-07-27T09:44:57.478147Z 6 [Warning] [MY-010453] [Server] root@localhost is created with an empty password ! Please consider switching off the --initialize-insecure option.
db-1  | 2026-07-27 09:44:59+00:00 [Note] [Entrypoint]: Database files initialized
db-1  | 2026-07-27 09:44:59+00:00 [Note] [Entrypoint]: Starting temporary server
web-1 | 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
web-1 | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
web-1 | /docker-entrypoint.sh: Configuration complete; ready for start up
web-1 | 2026/07/27 09:44:56 [notice] 1#1: using the "epoll" event method
web-1 | 2026/07/27 09:44:56 [notice] 1#1: nginx/1.31.3
db-1  | 2026-07-27T09:45:00.011373Z 0 [Warning] [MY-011068] [Server] The syntax '--skip-host-cache' is deprecated and will be removed in a future release. Please use SET GLOBAL host_cache_size=0 instead.
web-1 | 2026/07/27 09:44:56 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
db-1  | 2026-07-27T09:45:00.012718Z 0 [System] [MY-010116] [Server] /usr/sbin/mysqld (mysqld 8.0.46) starting as process 124
db-1  | 2026-07-27T09:45:00.032573Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
web-1 | 2026/07/27 09:44:56 [notice] 1#1: OS: Linux 6.18.33.2-microsoft-standard-WSL2
web-1 | 2026/07/27 09:44:56 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
web-1 | 2026/07/27 09:44:56 [notice] 1#1: start worker processes
web-1 | 2026/07/27 09:44:56 [notice] 1#1: start worker process 29
web-1 | 2026/07/27 09:44:56 [notice] 1#1: start worker process 30
web-1 | 2026/07/27 09:44:56 [notice] 1#1: start worker process 31
web-1 | 2026/07/27 09:44:56 [notice] 1#1: start worker process 32
web-1 | 2026/07/27 09:44:56 [notice] 1#1: start worker process 33
```



Conclusion:

This experiment demonstrated how Docker Compose simplifies the deployment and management of multi-service applications using a single configuration file. By creating and configuring a compose.yml file, multiple services such as an Nginx web server and a MySQL database were defined and deployed together. Docker Compose automatically created the required containers, networks, and volumes, allowing the services to communicate and run as a complete application stack.

During the deployment process, we used various Docker Compose commands to start, monitor, view logs, and remove services. A port conflict issue was encountered while starting the web service, which helped us understand the importance of proper configuration and troubleshooting in containerized environments.

Through this experiment, we learned how to create a multi-service application using Docker Compose, write and configure a Compose YAML file, deploy and manage containers using Docker Compose commands, monitor service status and logs, and remove containers, networks, and volumes when required. We also learned how to identify and resolve common deployment issues such as port conflicts. Overall, Docker Compose provides an efficient and organized method for managing applications with multiple interconnected services, making the deployment process faster, consistent, and easier to maintain.